(h)	**Synthetic polymers**
Students should:	
4.44	know that an addition polymer is formed by joining up many small molecules called monomers
4.45	understand how to draw the repeat unit of an addition polymer, including poly(ethene), poly(propene), poly(chloroethene) and (poly)tetrafluoroethene
4.46	understand how to deduce the structure of a monomer from the repeat unit of an addition polymer and vice versa
4.47	explain problems in the disposal of addition polymers, including: • their inertness and inability to biodegrade • the production of toxic gases when they are burned.
4.48C	**know that condensation polymerisation, in which a dicarboxylic acid reacts with a diol, produces a polyester and water**
4.49C	**understand how to write the structural and displayed formula of a polyester, showing the repeat unit, given the formulae of the monomers from which it is formed including the reaction of ethanedioic acid and ethanediol:** $n\,H-O-\overset{\displaystyle O}{\overset{\|}{C}}-\overset{\displaystyle O}{\overset{\|}{C}}-O-H + n\,H-O-CH_2CH_2-O-H \longrightarrow \left[-\overset{\displaystyle O}{\overset{\|}{C}}-\overset{\displaystyle O}{\overset{\|}{C}}-O-CH_2CH_2-O-\right]_n + 2n\,H_2O$
4.50C	**know that some polyesters, known as biopolyesters, are biodegradable**